20) An ovule which becomes curved so that the nucellus and embryo sac lie at right angles to the funicle is (AIPMT 2004)
a) camylotropous b) anatropous
c) orthotropous d) **hemianatropous**

21) Endosperm is formed during the double fertilization by (AIPMT 2000)
a) **two polar nuclei and one male gamete**
b) one polar nuclei and one male gamete
c) ovum and male gametes
d) two polar nuclei and two male gametes

UNIT VII – Genetics

1. Genes for cytoplasmic male sterility in plants are generally located in (AIPMT 2005)
a) **Mitrochondrial genome** b) Cytosol
c) Chloroplast genome d) Nuclear genome

2. In which mode of inheritance do you expect more maternal influence among the off spring (AIPMT 2006)
a) Autosomal b) **Cytoplasmic**
c) Y-linked d) X-linked

3. Which one of the following cannot be explained on the basis of Mendel's Law of Dominance? (AIPMT 2010)
a) Factors occur in pairs
b) The discrete unit controlling a particular character is called a factor
c) Out of one pair of factors one is dominant and the other is recessive
d) Alleles does not show any blending and both the characters recover as such in F_2 generation

4. F_2 generation in a Mendelian cross shows that both genotypic and phenotypic ratios are same as 1:2:1. It represents a case of (AIPMT 2012)
a) Monohybrid crosses with incomplete dominance
b) Co-dominance c) Dihybrid cross
d) Monohybrid cross with complete dominance

5. A Pleiotropic gene (AIPMT 2015 – Re-exam)
a) Controls multiple traits in an individual
b) Is expressed only in primitive plants
c) Is a gene evolved during Pliocene
d) Controls a trait only in combination with another L gene

6. A true breeding plant is (NEET Phase II 2016)
a) Near homozygous and produces offspring of its own kind
b) Always homozygous recessive in its genetic construction
c) One that is able to breed on its own
d) Produced due to cross pollination among unrelated plants

7. Mendel obtained wrinkled seeds in pea due to the deposition of sugars instead of starch. It was due to which enzyme? (AIPMT 2001)
a) Amylase b) Invertase c) Diastase
d) **Absence of starch branching enzyme**

8. Ratio of complementary gene is (AIPMT 2001)
a) 9:3:4 b) 12:3:1 c) 9:3:3:4 d) **9:7**

9. If there are 999 bases in an RNA that codes for a protein with 333 amino acid and the base at position 901 is deleted such that the length of the RNA becomes 998 bases, how many codons will be altered? (NEET 2017)
a) 1 b) 11 c) 33 d) **333**

10. If a homozygous red flowered plant is crossed with a homozygous white flowered plant, then the off-springs will be (AIIMS 1999, 2002, 2007)
a) Half-white flowered b) Half-red flowered
c) All white flowered d) **All red flowered**

11. The ratio in a dihyrbid test cross between two individuals is given by (AIIMS 2001)

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